

Các loại query trong JPA

1 JPQL cơ bản (SELECT)

```
// Lấy tất cả User
List<User> list = em.createQuery(
    "SELECT u FROM User u", User.class
).getResultList();
```

Ý nghĩa:

- JPQL thao tác trên entity (User), không phải bảng DB

2 WHERE + tham số

```
// Tìm theo email
User user = em.createQuery(
    "SELECT u FROM User u WHERE u.email = :email", User.class
)
.setParameter("email", "a@gmail.com")
.getSingleResult();
```

=> Luôn dùng parameter để tránh SQL Injection

3 LIKE (tìm kiếm)

```
List<User> list = em.createQuery(
    "SELECT u FROM User u WHERE u.name LIKE :kw", User.class
)
.setParameter("kw", "%Nguyen%")
.getResultList();
```

4 ORDER BY (sắp xếp)

```
List<User> list = em.createQuery(
    "SELECT u FROM User u ORDER BY u.name ASC", User.class
).getResultList();
```

5 COUNT

```
long total = em.createQuery(
    "SELECT COUNT(u) FROM User u", Long.class
).getSingleResult();
```

6 JOIN (liên kết entity)

Giả sử:

- User có nhiều Order

```
// Lấy user và order
List<Object[]> list = em.createQuery(
    "SELECT u, o FROM User u JOIN u.orders o", Object[].class
).getResultList();
```

Hoặc:

```
// Lấy user có order
List<User> list = em.createQuery(
    "SELECT DISTINCT u FROM User u JOIN u.orders o", User.class
).getResultList();
```

7 Named Query (khai báo sẵn trong Entity)

Trong entity:

```
@NamedQuery(
    name = "User.findByEmail",
    query = "SELECT u FROM User u WHERE u.email = :email"
)
@Entity
public class User { ... }
```

Sử dụng:

```
User user = em.createNamedQuery("User.findByEmail", User.class)
    .setParameter("email", "a@gmail.com")
    .getSingleResult();
```

8 Native Query (SQL thuần)

```
List<Object[]> list = em.createNativeQuery(
    "SELECT * FROM users WHERE email = ?"
)
    .setParameter(1, "a@gmail.com")
    .getResultList();
```

Hoặc map về entity:

```
List<User> list = em.createNativeQuery(
    "SELECT * FROM users", User.class
).getResultList();
```

9 UPDATE (JPQL)

```
int updated = em.createQuery(
    "UPDATE User u SET u.name = :name WHERE u.id = :id"
)
    .setParameter("name", "New Name")
```

```
.setParameter("id", 1)
.executeUpdate();
```

Lưu ý:

- Phải nằm trong transaction

10 DELETE (JPQL)

```
int deleted = em.createQuery(
    "DELETE FROM User u WHERE u.id = :id"
)
.setParameter("id", 1)
.executeUpdate();
```

11 Criteria API (type-safe, nâng cao)

```
CriteriaBuilder cb = em.getCriteriaBuilder();
CriteriaQuery<User> cq = cb.createQuery(User.class);

Root<User> root = cq.from(User.class);

cq.select(root)
    .where(cb.equal(root.get("email"), "a@gmail.com"));

List<User> list = em.createQuery(cq).getResultList();
```

12 Dynamic query (kết hợp nhiều điều kiện)

```
String jpql = "SELECT u FROM User u WHERE 1=1";

if (name != null) {
    jpql += " AND u.name LIKE :name";
}

TypedQuery<User> query = em.createQuery(jpql, User.class);

if (name != null) {
    query.setParameter("name", "%" + name + "%");
}

List<User> list = query.getResultList();
```